



Praktikum Design of Wind Farms

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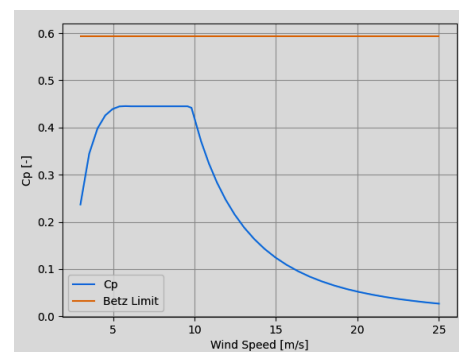
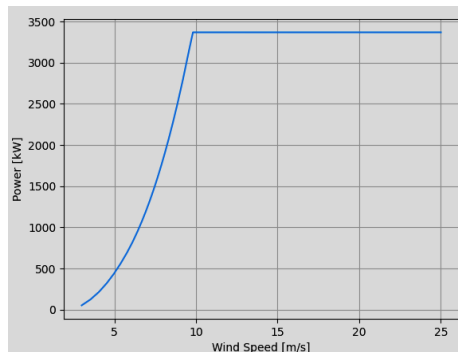
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Assignment 3 (8%): Wind Farm Analysis Using FLORIS

This assignment aims at introducing students to the use of FLORIS including defining a custom wind turbine, checking and plotting the turbine characteristics, defining a wind farm in FLORIS, and calculating the power output of the wind farm for different wind speeds and directions using specific wake models.

Three IEA 3.4MW wind turbines are planned to be installed in a row (north to south) with a separation distance of 5D between each successive turbine. The wind shear exponent and TI are 0.14 and 6%, respectively. The key parameters of the turbines and their characteristics are as follows:

Item	Value	Units
Name	IEA 3.4MW Reference	N/A
Rated Power	3370	kW
Rated Wind Speed	9.8	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	130	m
Hub Height	120	m
Control	Pitch Regulated	N/A



A .csv file is available on [GitHub](#).

1. Define the IEA3.4 MW wind turbine in FLORIS and add it to the “turbine_library” folder.
2. Check the turbine characteristics Ct-curve, Cp-curve, Power-curve, and power vs. yaw angle $[-30^\circ, +30^\circ]$ by plotting them for air densities of 1.150 and 1.225 (ref) and wind speed of 7.5 m/s for yaw error testing (Hint: in Floris v3, check “check-turbine.py”, in Floris v4, check “examples_turbines\001_check_turbine.py” in the example files).
3. Define the wind farm in FLORIS and visualize the wind farm flow field in the (a) horizontal plane at hub height, (b) stream-wise plane, and (c) span-wise plane when the wind speed is 7.5 m/s for two wind directions of 0° and 240° (Hint: in Floris v3, check



“visualizations.py”, in Floris v4, check “002_visualizations.py” and the folder “examples_visualizations”).

4. Use Jensen’s wake model ($k_w = 0.075$), and determine the output power of each wind turbine as well as the output power of the farm when the wind speed is 7.5 m/s for two wind directions of 0° and 240° (Hint: in Floris v3, check “opening_floris_computing_power.py”, in Floris v4 check “005_getting_power.py” in the example files). Compare the results with the power output of the farm calculated in Assignment 2. Calculate the relative errors (percentage) with respect to the results of Assignment 2. Name at least one reason for the relative error.
5. Repeat question 4 only for a wind direction of 0° and if the separation distance increases to 10D. Calculate the relative difference (percentage) of the output power of the farm with respect to the case of 5D.
6. As you might know, the atmospheric boundary layer stability, which is changing during the day (and year), affects the wind characteristics such as wind shear and turbulence intensity. If during a winter season, the wind shear exponent and turbulence intensity change as shown in the following table:

	Wind shear exponent	Turbulence intensity
Day	0.06	12%
Night	0.22	6%

Use the Gaussian wake model and determine the output power of each wind turbine and visualize the flow field for night and day conditions when the wind speed is 7.5 m/s, wind direction is 0° and the separations distance is 5D. Explain your answer and compare the results and recovery of the turbine wakes for day and night. How do these parameters affect power generation?

Note: Please upload your code and a Word/PDF file in which you describe the steps of your solution and show/explain your results. It is not enough to only upload code files.